BOAMP Procurement Renewal Linking

Final Technical Report — Phase 1

Data Audit, Preprocessing, Mathematical Formulation, and Renewal Event Construction

Internship Project — Gigalis

June 2026

Abstract

This report presents the final Phase 1 methodology for reconstructing renewal links in BOAMP public procurement data. BOAMP notices do not contain an explicit field identifying whether a call-for-tender (*appel d'offre*) renews a previous contract. The renewal relationship must therefore be inferred from observable signals: buyer identity, contract description, CPV classification, and timing relative to the expected contract end date. Starting from 1,933 *APPEL_OFFRE* notices over 2015–2024, the pipeline defines 1,100 eligible source contracts and links 697 of them to a later renewal candidate, giving a final linking rate of 63.4%. The previous baseline approach linked 219 notices out of all 1,933 *APPEL_OFFRE* notices (11.3%); the final method uses the narrower denominator of 1,100 notices whose expected renewal date is observable within the study period. The final output is `boamp_renewal_links.csv`, a modeling table with one row per eligible source contract and an `event` variable for Phase 2.

Executive Summary

The objective of Phase 1 is not to prove with legal certainty that two BOAMP notices are renewals of one another. The objective is to build a transparent, auditable, and reproducible proxy event variable for survival modeling. The final algorithm uses four signals: semantic similarity of the contract description, CPV compatibility, temporal proximity to the expected renewal date, and buyer identity.

The final method links 697 out of 1,100 eligible source contracts, or 63.4%. The gain comes mainly from two methodological changes: using Sentence-Transformer embeddings instead of lexical Jaccard/TF-IDF matching, and treating CPV as a scored compatibility signal rather than as a hard grouping constraint. The largest remaining weakness is buyer identification: only 5.9% of buyers are identified by SIRET, so most grouping relies on normalised buyer names. This can fragment the same public entity into several buyer keys and create false negatives.

The `event` variable should be interpreted carefully. `event = 1` means that the algorithm found a plausible renewal candidate. `event = 0` means that no candidate passed the linking rule in the observed data. It should not be read as proof that the contract was never renewed.

Contents

1 Context and Objective

1.1 Business Context

Gigalis manages a portfolio of public digital infrastructure contracts in the Pays-de-la-Loire region. A critical operational question is: *when is a procurement contract likely to be renewed?* Knowing this in advance allows account managers to anticipate renewal campaigns, prepare competitive offers, and reduce the risk of losing contracts by default.

1.2 Why BOAMP?

The **Bulletin Officiel des Annonces de Marchés Publics** (BOAMP) is the official French public procurement gazette. All public contracts above the relevant thresholds must be published there, making it the most complete source of historical procurement data in France. It covers the full lifecycle: pre-information, call for tender (*appel d'offre*), award (*attribution*), and modification notices.

1.3 Core Problem

BOAMP does **not** provide an explicit field linking an original call for tender to its renewal. Each notice is a standalone record. The renewal relationship must therefore be **inferred** from a combination of signals: the same buyer re-issuing a contract for a similar service in the expected time-window after the original contract was due to expire.

1.4 Scope of Phase 1

Phase 1 produces a single output table: one row per eligible *APPEL_OFFRE* with a binary column `event`. In this report, `event = 1` means that a plausible renewal was found by the linking algorithm, while `event = 0` means that no renewal link was observed under the chosen rule. This table is the direct input to Phase 2, where the event definition and censoring treatment must be made explicit in the survival model.

2 Raw Dataset

2.1 Source and Extraction

The dataset was extracted from the BOAMP API and covers notices published from **2015-03-02 to 2024-12-29**. The raw file after initial cleaning is `data/processed/boamp_full_clean.csv`.

The renewal linking algorithm operates exclusively on the 1,933 *APPEL_OFFRE* notices. ATTRIBUTION notices are used only as a secondary source to extract more precise contract start dates (via the `annonce_lie` back-reference field).

Table 1: Dataset overview

Notice type	Count	Share
APPEL_OFFRE (calls for tender)	**1,933**	**60.8%**
ATTRIBUTION (award notices)	1,081	34.0%
RECTIFICATIF (corrections)	150	4.7%
PRE-INFORMATION	7	0.2%
MODIFICATION	5	0.2%
INTENTION_CONCLURE	4	0.1%
PERIODIQUE	1	0.0%
Total	**3,181**	100%

2.2 Column Dictionary — Raw Dataset (44 columns)

Table 2 describes all 44 columns in `boamp_full_clean.csv`. Columns are grouped by origin: raw BOAMP fields and fields added by the cleaning pipeline.

Table 2: Full column dictionary — `boamp_full_clean.csv`

Column	**Type**	**Coverage**	**Description**
— *Identifiers and dates* —			
`idweb`	string	100%	Unique BOAMP notice identifier (e.g. `22-111359`). Primary key.
`dateparution`	date	100%	Official publication date of the notice. Used as contract start proxy when no award date exists.
`datefindiffusion`	date	100%	End-of-diffusion date (administrative). Not used in modelling.
`datelimitereponse`	datetime	42.8%	Deadline for tender submission. Only partially filled.
`date_attribution`	date	0%*	Award date parsed from the notice body (0% on AO records; populated only on ATTRIBUTION notices).
`date_publication_anterieure`	date	0.4%	Reference to a prior publication. Sparse; not used.
— *Buyer information* —			
`nomacheteur`	string	100%	Raw buyer name as published. Inconsistent casing and accents.
`code_departement`	string	100%	French department code (e.g. `44` for Loire-Atlantique).
`buyer_siret`	string	9.9%	SIRET number (14-digit unique company ID). Low coverage.
`buyer_cp`	string	96.5%	Postal code of the buyer.
`buyer_ville`	string	100%	City name of the buyer.
`buyer_key`†	string	100%	**Constructed key**: `SIRET:XXXXXXXX` when SIRET is available, else `NAME:normalised_name`. Used as the grouping key for renewal matching.
— *Contract classification* —			
`famille`	string	100%	BOAMP family code (`JOUE` = EU Journal, or other).
`nature`	string	100%	Notice type (see Table 1 above).
`nature_libelle`	string	100%	Human-readable notice type label.
`type_procedure`	string	98.5%	Procurement procedure: OUVERT (62.4%), PROCEDURE_ADAPTE (24.3%), NEGOCIE (8.9%), RESTREINT (1.9%), other.
`type_marche`	string	91.6%	Contract type: SERVICES (53.6%), FOURNITURES (26.2%), TRAVAUX (12.0%), mixed.
`type_avis`	string	100%	Notice subtype codes (e.g. `5|1|`).

Continued on next page

Table 2 (continued)

Column	Type	Coverage	Description
etat	string	100%	Publication state (INITIAL 97.6%, ANNULE, RECTIFIE).
contractfolderid	string	9.8%	Internal contract folder ID. Very sparse; not used.
— *Contract description* —			
objet	string	100%	Free-text contract description. Mean length: 105 characters (range: 6–201). The primary text signal for semantic matching.
descripteur_libelle	string	100%	Thematic descriptor in natural language (e.g. "Informatique (prestations de services)"). 549 unique values.
titulaire	string	0%	Winning contractor (populated only on ATTRIBUTION, not AO).
annonce_lie	string	2.9%	Back-reference to an earlier notice. On ATTRIBUTION records: 82.9% filled. Used as partial ground truth for calibration.
— *CPV codes* —			
cpv_principal	float	96.8%	Raw CPV code (8-digit integer) as published.
cpv_all	string	100%	Pipe-delimited list of all CPV codes for the notice.
cpv_clean†	float	96.8%	Cleaned primary CPV code. Generic codes (ending in 000000) flagged separately.
cpv_div2†	float	96.8%	First 2 digits of CPV (division-level, e.g. 72 = IT services). Range: 15–98.
cpv_class4†	float	96.8%	First 4 digits of CPV (class-level granularity).
category_id†	string	100%	Internal category code (CAT01–CAT11) derived from CPV.
category_label†	string	100%	Human-readable category (see Table 3).
— *Contract amount* —			
amount_eur	float	22.4%	Raw amount in euros as parsed from the notice.
amount_raw	float	22.4%	Amount before cleaning transformations.
amount_clean†	float	21.4%	Cleaned amount: zero-values, suspiciously small values, and ceiling-flagged amounts removed. Range: €2,000 – €9.6M; median €350,000.
flag_amount_zero†	int	100%	1 if amount_eur = 0 (reported but missing).
flag_amount_tiny†	int	100%	1 if amount < €10,000 (likely misreported).
flag_amount_ceiling†	int	100%	1 if amount > €960,000 (likely a framework ceiling, not actual spend).
— *Contract duration* —			
duration_months	float	76.9%	Duration in months as parsed from the notice.
duration_raw_boamp	float	76.9%	Raw parsed value before cleaning.
duration_source_field	string	76.9%	Source field from which duration was extracted (DUREE_MOIS, DUREE_JOURS, etc.).
flag_duration_suspect†	int	100%	1 if duration > 120 months (10 years) or = 0.
duration_clean†	float	76.6%	Cleaned duration: suspect values set to missing. Range: 1–120 months; median 24 months (among non-imputed).
— *Other* —			
url_avis	string	100%	Direct URL to the notice on boamp.fr.
donnees_format	string	100%	Data format flag (legacy 87.3%, json 12.7%).

† Column added or derived by the cleaning pipeline. Not present in the original BOAMP API response.
* date_attribution is 0% for AO notices; it is populated on ATTRIBUTION notices and used indirectly via annonce_lie.

2.3 Contract Category Distribution

The 11 internal categories are derived from the CPV 2-digit division and cover exclusively the ICT/digital services scope of the Gigalis portfolio.

Table 3: Category distribution among APPEL_OFFRE notices

Category ID	Label	Count	Share
CAT02	IT Services & Consulting	312	16.1%
CAT01	Software & Applications	209	10.8%
CAT03	Telecom & Networks	204	10.6%
CAT04	Cybersecurity	121	6.3%
—	Unknown (CPV missing)	108	5.6%
CAT05	Digital Workplace & Collab.	45	2.3%
CAT06	Data & AI	34	1.8%
CAT07	IT Maintenance & Support	22	1.1%
CAT08	IT Hardware & Equipment	21	1.1%
CAT09	Cloud & Infrastructure	19	1.0%
CAT10	GIS & Mapping	5	0.3%
	Total AO	**1,933**	100%

2.4 Buyer Landscape

Table 4: Buyer concentration among APPEL_OFFRE notices

Metric	Value
Total unique buyers (`buyer_key`)	475
Buyers identified by SIRET	28 (5.9%)
Buyers identified by normalised name only	447 (94.1%)
Buyers with ≥ 2 AO notices	225 (47.4%)
Buyers with ≥ 5 AO notices	78 (16.4%)
Top 5 buyers by number of AO notices	
Nantes Métropole	128
Des Pays de la Loire	109
De Nantes	61
Saint-Nazaire	50
Angers Loire Métropole	46

The dominance of name-based buyer keys (94.1%) is the primary structural risk: variant spellings of the same buyer (e.g. “Nantes” vs. “Nantes Métropole” vs. “NM”) fragment the buyer group, creating structural misses that the algorithm cannot recover.

3 Data Quality Issues Diagnosed

Nine data quality issues were identified through systematic EDA. Each directly affects the accuracy or recall of the renewal linking algorithm.

4 Preprocessing Pipeline

4.1 Buyer Key Construction

$$\texttt{buyer\_key}_i = \begin{cases} \texttt{SIRET:}s_i & \text{if SIRET } s_i \text{ is available} \\ \texttt{NAME:}\phi(n_i) & \text{otherwise} \end{cases} \tag{1}$$

where $\phi(\cdot)$ is the name normalisation function: lowercase, strip accents, remove punctuation, collapse whitespace.

4.2 Contract Start Date Refinement

Award date from linked ATTRIBUTION notices is used when available, falling back to the publication date:

$$t_i = \begin{cases} \text{award\_date}(j) & \text{if } \exists\, j \in \text{ATTR} : \texttt{annonce\_lie}(j) = i \\ \texttt{dateparution}_i & \text{otherwise} \end{cases} \tag{2}$$

In practice, 40.7% of eligible AO obtain a more precise start date through this refinement.

4.3 Duration Imputation

$$d_i = \begin{cases} \texttt{duration\_clean}_i & \text{if not missing and not suspect} \\ d_0 = 48 \text{ months} & \text{otherwise (23.4\% of AO)} \end{cases} \tag{3}$$

4.4 Estimated End Date

$$\hat{e}_i = t_i + d_i \tag{4}$$

4.5 Text Normalisation

The *objet* field is normalised for TF-IDF (used in calibration) and passed raw to Sentence-Transformers:

1. Lowercase
2. Strip Unicode accents (NFD normalisation)
3. Keep only characters matching `[a-z0-9 ]`
4. Collapse multiple spaces

4.6 Sentence-Transformer Embeddings

Each normalised *objet* $\mathbf{o}_i$ is encoded into a 384-dimensional unit-norm vector:

$$\mathbf{v}_i = \mathrm{ST}\big(\mathbf{o}_i\big) \in \mathbb{R}^{384}, \qquad \|\mathbf{v}_i\|_2 = 1 \tag{5}$$

Model used: `paraphrase-multilingual-MiniLM-L12-v2` (multilingual, 384 dimensions, ~120 MB).

5 Renewal Linking Algorithm

5.1 Notation

Let $\mathcal{A}$ denote the set of all *APPEL_OFFRE* notices. Each notice $i \in \mathcal{A}$ is represented by:

- t_i: contract start date, defined as the award date when available, otherwise the BOAMP publication date;
- d_i: contract duration in months, after cleaning and imputation;
- $\hat{e}_i = t_i + d_i$: estimated contract end date;
- CPV_i: main CPV classification code;
- $\mathbf{o}_i$: contract description text (*objet*);
- b_i: buyer identifier, using SIRET when available and normalised buyer name otherwise.

The goal is to construct, for each eligible source notice i, a binary outcome Y_i indicating whether a plausible renewal candidate can be found among later BOAMP calls-for-tender.

5.2 Step 1 — Eligibility Filter

A notice is **eligible** to be a renewal source if its estimated renewal date falls within the study period $[T_{\min}, T_{\max}]$:

$$i \in \mathcal{A}_{\text{elig}} \iff T_{\min} \le \hat{e}_i \le T_{\max} \quad \big(T_{\min} = \text{2015-01-01},\ T_{\max} = \text{2024-12-31}\big) \tag{6}$$

Contracts with $\hat{e}_i > T_{\max}$ are excluded from the Phase 1 linking denominator because the renewal opportunity is not sufficiently observable in the 2015–2024 dataset.

5.3 Step 2 — Candidate Pool

For each eligible source i, the candidate pool is:

$$\mathcal{C}(i) = \{ j \in \mathcal{A} : b_j = b_i, \quad t_j > t_i, \quad |t_j - \hat{e}_i| \leq W \}, \quad W = 12 \text{ months} \tag{7}$$

The gap between source and candidate is:

$$\delta(i, j) = t_j - t_i \quad \text{(in months)} \tag{8}$$

Key design choice: candidates are grouped by `buyer_key` only. The previous baseline grouped by `[buyer_key, cpv_div2]`, which rejected valid renewals where the buyer reclassified the service under a different CPV code.

5.4 Step 3 — Component Scores

Text similarity. Cosine similarity between unit-norm embeddings (dot product = cosine when both vectors are ℓ_2-normalised):

$$s_{\text{text}}(i, j) = \mathbf{v}_i \cdot \mathbf{v}_j \in [0, 1] \tag{9}$$

CPV compatibility. A 4-level step function. A CPV code is treated as generic if it ends in `000000`, for example `72000000` for generic IT services.

$$s_{\text{cpv}}(i, j) = \begin{cases} 0.5 & \text{either CPV missing} \\ 0.5 & \text{either generic \& same 2-digit div.} \\ 0.0 & \text{either generic \& different div.} \\ 1.0 & \text{CPV}[:4]_i = \text{CPV}[:4]_j \text{ (same class)} \\ 0.7 & \text{CPV}[:2]_i = \text{CPV}[:2]_j \text{ (same division)} \\ 0.3 & \text{CPV}[:1]_i = \text{CPV}[:1]_j \text{ (same section)} \\ 0.0 & \text{otherwise} \end{cases} \tag{10}$$

Temporal proximity. Linear decay from 1.0 at the exact expected renewal date to 0.0 at the window boundary $W = 12$ months:

$$s_{\text{temp}}(i, j) = \max\left(0,\ 1 - \frac{|\delta(i, j) - d_i|}{W}\right) \tag{11}$$

Buyer match. Always 1.0 within the same-buyer grouping loop:

$$s_{\text{buyer}}(i, j) = 1.0 \tag{12}$$

5.5 Step 4 — Composite Score

$$\boxed{S(i,j) = 0.40 \cdot s_{\text{text}} + 0.25 \cdot s_{\text{cpv}} + 0.20 \cdot s_{\text{temp}} + 0.15 \cdot s_{\text{buyer}}} \tag{13}$$

5.6 Step 5 — Hard Filters

Two hard gates are applied before best-match selection:

$$s_{\text{text}}(i,j) \geq 0.20 \tag{14}$$
$$\delta(i,j) \in [6,\ 72] \text{ months} \tag{15}$$

The minimum gap of 6 months prevents matching a contract to itself or to an immediate amendment. The maximum gap of 72 months (6 years) covers contracts with the longest expected durations.

CPV is intentionally **not** used as a hard filter. Requiring CPV agreement as a gate would exclude legitimate renewals when the same buyer reclassifies a similar service under a different CPV code.

Let $\mathcal{F}(i)$ denote the filtered candidate set:

$$\mathcal{F}(i) = \big\{ j \in \mathcal{C}(i) : s_{\text{text}}(i,j) \geq 0.20,\ \delta(i,j) \in [6,72] \big\}. \tag{16}$$

5.7 Step 6 — Best-Match Selection

For each source, keep the single highest-scoring candidate:

$$j^*(i) = \arg\max_{j \in \mathcal{F}(i)} \ S(i,j) \tag{17}$$

where $\mathcal{F}(i) \subseteq \mathcal{C}(i)$ is the subset passing both hard filters (14) and (15).

5.8 Step 7 — Outcome Variable

$$Y_i = \begin{cases} 1 & \text{if } \mathcal{F}(i) \neq \emptyset \quad \text{(renewal found)} \\ 0 & \text{if } \mathcal{F}(i) = \emptyset \quad \text{(no renewal link observed)} \end{cases} \tag{18}$$

The final linking rate is:

$$\text{Linking rate} = \frac{\sum_{i \in \mathcal{A}_{\text{elig}}} Y_i}{|\mathcal{A}_{\text{elig}}|} = \frac{697}{1{,}100} = 63.4\%. \tag{19}$$

The value $Y_i = 0$ should be interpreted as an unobserved link under the available data and chosen thresholds, not as definitive evidence that no renewal occurred.

6 Output Dataset — Modeling Input

6.1 File: `boamp_renewal_links.csv`

This is the direct input to the survival model. One row per eligible *APPEL_OFFRE* (1,100 rows, 17 columns).

Table 8: Column dictionary — `boamp_renewal_links.csv` (modeling input)

Column	Type	Description
— *Identifiers* —		
`contract_id`	string	Source contract BOAMP ID (e.g. `BOAMP:15-31464`).
`renewal_contract_id`	string	Best-matched renewal contract ID. Empty when `event=0`.
— *Buyer and classification* —		
`buyer_key`	string	Buyer identifier (`SIRET:...` or `NAME:...`). 308 unique buyers.
`cpv_div2`	float	2-digit CPV division of the source contract. 95.3% coverage.
`category_label`	string	Internal category (11 levels; see Table 3).
— *Temporal variables* —		
`contract_start`	date	Source contract start date (refined from award date where available).
`declared_duration_months`	float	Contract duration as declared (or imputed). Mean: 31.5 months.
`dur_was_imputed`	bool	True if duration was set to the default 48-month imputation (25.1% of rows).
`estimated_end_date`	date	$\hat{e}_i = t_i + d_i$; defines the renewal search window.
`observed_duration_months`	float	Actual gap $\delta(i, j^*)$ in months for linked contracts. Range: 6–71.5. Empty when `event=0`.
— *Outcome variable* —		

Table 8 (continued)

Column	Type	Description
`event`	int	**Target variable**. 1 = plausible renewal link found (697), 0 = no renewal link observed under the algorithm (403).
— *Matching scores (linked rows only)* —		
`composite_score`	float	$S(i, j^*)$. Mean: 0.624, range: 0.251–0.998.
`text_similarity`	float	$s_{\text{text}}(i, j^*)$. Mean: 0.513, range: 0.200–1.000.
`cpv_match_score`	float	$s_{\text{cpv}}(i, j^*)$. Mean: 0.527, range: 0.000–1.000.
`temporal_score`	float	$s_{\text{temp}}(i, j^*)$. Mean: 0.684, range: 0.007–1.000.
— *Metadata* —		
`start_date_source`	string	How t_i was obtained: `publication_date` (59.3%) or `attribution_award_date` (40.7%).
`link_method`	string	Algorithm used: `sentence_transformers` (697 rows) or `none` (403 rows).

6.2 Score Distributions for Linked Contracts

Table 9 reports the distribution of matching scores among the 697 linked contracts. These statistics are useful diagnostics, but they should not be interpreted as independent validation: the same scores are used by the algorithm to select the links. The low minimum values, especially for the composite, CPV, and temporal scores, indicate that some accepted links are low-confidence cases that should be checked in the Phase 2 validation sample.

Table 9: Descriptive statistics of matching scores (event=1 rows only, n=697)

Score	Mean	Std	Min	Q25	Median	Q75	Max
`composite_score`	0.624	0.143	0.251	0.520	0.622	0.716	0.998
`text_similarity`	0.513	0.202	0.200	0.360	0.475	0.619	1.000
`cpv_match_score`	0.527	0.353	0.000	0.300	0.500	0.700	1.000
`temporal_score`	0.684	0.255	0.007	0.541	0.758	0.896	1.000

6.3 Declared vs. Observed Duration

Table 10 compares the declared contract duration with the gap between the source notice and the selected renewal candidate. This is a consistency check for the temporal matching logic, not external proof that the selected pairs are true renewals, because declared duration is already used to define the candidate window and the temporal score.

Table 10: Duration statistics (months)

	Mean	Std	Min	Q25	Median	Q75	Max
`declared_duration_months` (all 1,100)	31.5	17.7	1	12	36	48	72
`observed_duration_months` (697 linked)	32.3	17.0	6	13	37	48	71

The close medians, 36 months for declared duration and 37 months for the linked candidate gap, are therefore reassuring but partly mechanical. A stronger validation should examine the error `observed_duration_months` − `declared_duration_months`, separately for imputed and non-imputed durations, and combine this with manual review of a stratified sample of linked pairs.

7 Results

7.1 Linking Rate Comparison

The two key changes responsible for the improvement are:

1. **Grouping by buyer only** (removing CPV as a hard gate): recovers contracts where the buyer reclassified the service. The final method also reports performance on the 1,100 contracts whose expected renewal date is observable within the study period, instead of using all 1,933 AO notices as the denominator.

2. **Sentence-Transformers over TF-IDF**: semantic similarity captures meaning beyond lexical overlap, contributing approximately +38 percentage points in linking rate compared to TF-IDF alone.

7.2 Failure Analysis

Of the 403 unlinked eligible AO, three failure modes are identified:

The dominant failure mode (`NO_TEMPORAL_PARTNER`, 84.1%) reflects a structural ceiling: buyers who published only one call-for-tender within the study period cannot be linked regardless of algorithm quality. This is the main remaining ceiling on the 36.6% unlinked fraction.

8 Worked Example

Table 13 illustrates the score computation for the first pair in `boamp_renewal_candidates.csv`.

This pair passes both hard filters ($s_{\text{text}} = 0.69 \geq 0.20$; gap $= 33.5 \in [6, 72]$ months) and is selected as the best match for source `15-31464`.

Concrete step

This part illustrates how the renewal-linking score is computed for one concrete candidate pair. The example comes from the first row of `boamp_renewal_candidates.csv`.

The source contract is:

$$\texttt{src} = \texttt{15-31464}$$

The candidate renewal notice is:

$$\texttt{cand} = \texttt{17-176560}$$

Both notices belong to the same buyer group:

$$\text{Buyer} = \text{Département de la Mayenne}$$

Source and Candidate Information

The source contract started on 2015-03-02 and had a declared duration of 36 months. Therefore, the expected renewal date is around March 2018:

$$\hat{e}_i = \text{2015-03-02} + 36 \text{ months} \approx \text{2018-03}$$

The candidate notice was published on 2017-12-17. This is close to the expected end date of the original contract, so it is a plausible renewal candidate.

Step 1: Time Gap

The time gap between the source contract and the candidate is:

$$\delta(i, j) = t_j - t_i$$

In this example:

$$\delta(i, j) = 2017\text{-}12\text{-}17 - 2015\text{-}03\text{-}02 = 33.54 \text{ months}$$

The declared duration was:

$$d_i = 36 \text{ months}$$

So the candidate appears:

$$36 - 33.54 = 2.46 \text{ months}$$

before the expected contract end date.

Step 2: Temporal Score

The temporal score is:

$$s_{\text{temp}}(i, j) = 1 - \frac{|\delta(i, j) - d_i|}{W}$$

with:

$$W = 12$$

Therefore:

$$s_{\text{temp}}(i, j) = 1 - \frac{|33.54 - 36|}{12}$$

$$s_{\text{temp}}(i, j) = 1 - \frac{2.46}{12}$$

$$s_{\text{temp}}(i, j) = 0.7951$$

This is a high temporal score because the candidate appears only about 2.5 months before the expected renewal date.

Step 3: Other Component Scores

The text similarity score is computed using the cosine similarity between the two Sentence-Transformer embeddings:

$$s_{\text{text}}(i,j) = 0.6922$$

This means that the two contract descriptions are semantically close.

The CPV score is:

$$s_{\text{cpv}}(i,j) = 0.5$$

because the source CPV code is missing. The algorithm gives a neutral score instead of rejecting the pair. This is important because missing CPV should not automatically imply that two notices are unrelated.

The buyer score is:

$$s_{\text{buyer}}(i,j) = 1.0$$

because the two notices belong to the same buyer group.

Step 4: Composite Score

The composite score is the weighted sum of the four component scores:

$$S(i,j) = 0.40 s_{\text{text}} + 0.25 s_{\text{cpv}} + 0.20 s_{\text{temp}} + 0.15 s_{\text{buyer}}$$

Substituting the values:

$$S(i,j) = 0.40(0.6922) + 0.25(0.5) + 0.20(0.7951) + 0.15(1.0)$$

$$S(i,j) = 0.2769 + 0.1250 + 0.1590 + 0.1500$$

$$S(i,j) = 0.7109$$

So the final composite score is:

$$\boxed{S(i,j) = 0.7109}$$

This matches the `composite_score` reported in the candidate file.

Interpretation

This pair is considered a plausible renewal link because all main signals are consistent:

- the buyer is the same;
- the candidate appears close to the expected contract end date;
- the contract descriptions are semantically similar;
- the CPV code is missing for the source, but this is treated neutrally rather than as evidence against the match.

The pair also passes the two hard filters:

$$s_{\text{text}} = 0.6922 \geq 0.20$$

and:

$$\delta(i, j) = 33.54 \in [6, 72]$$

Therefore, this candidate remains in the filtered candidate set. If it has the highest composite score among all surviving candidates for source `15-31464`, it is selected as the best renewal match:

$$j^*(i) = \texttt{17-176560}$$

This example shows exactly how the algorithm transforms raw BOAMP notices into a renewal-linking score. The score is not legal proof of renewal. It is an algorithmic measure of plausibility based on buyer identity, timing, text similarity, and CPV compatibility.

9 Limitations and Risks

L1. No explicit ground truth. BOAMP does not provide a renewal field. The `annonce_lie` signal on ATTRIBUTION notices calibrates text similarity thresholds but is not a direct renewal label. The true precision of the 697 links is unknown without manual validation.

L2. Name-based buyer fragmentation. 94.2% of buyer keys are name-based. Variant spellings of the same public entity (abbreviations, accents, word order) create separate groups, producing structural false negatives that cannot be recovered by tuning thresholds alone.

L3. Duration imputation introduces noise. 25.1% of eligible AO had their duration imputed to 48 months. If the true duration differs significantly, the temporal window is shifted and the correct renewal candidate may fall outside the ±12-month window.

L4. False positives in sparse-buyer groups. Relaxing CPV from a hard filter to a scored component increases recall but also increases the risk of linking two unrelated contracts from the same buyer that happen to use similar procurement language (e.g. generic IT services descriptions).

L5. Sample scope. The dataset covers only the Pays-de-la-Loire region and is focused on ICT/digital services. Linking rates and score distributions may differ significantly for other regions or sectors.

Table 5: Data quality issues and their impact on the linking algorithm

#	Issue	Observation	Impact on linking
1	SIRET coverage low	Only 5.8% of AO have a SIRET; the rest use normalised name as buyer key	Variant spellings split buyer groups ⇒ structural misses
2	Buyer name variants	Same buyer appears under multiple spellings after normalisation	False negative links where buyers are split
3	CPV missing or generic	3.2% of AO have no CPV; no records use a generic catch-all code	CPV score defaults to neutral (0.5) for those pairs
4	Contract amount sparse	Only 21.4% of AO report a valid, non-flagged amount	Amount cannot be used as a matching signal
5	Duration missing	23.4% of AO have no duration; imputed to 48 months	Wrong imputed duration shifts the temporal window
6	Duration suspect values	Some raw durations > 120 months or = 0	Removed; fall into imputation bucket
7	Contract start date imprecise	`date_attribution` is 0% on AO records; publication date used as proxy for 59.3%	Up to several weeks of noise in temporal scoring
8	Short or generic *objet*	24 notices have `objet` < 20 characters	Semantic embeddings less discriminative for short texts
9	`annonce_lie` interpretation	Present only on ATTRIBUTION notices (82.9%); refers to same-contract back-link, not a renewal	Used as calibration signal, not as a direct renewal label

Table 6: Eligibility breakdown

Population	Count
Total APPEL_OFFRE	1,933
Not eligible: estimated end date after 2024	833
Eligible ($\mathcal{A}_{\text{elig}}$)	**1,100**

Table 7: Weight rationale

Component	Weight	Rationale
s_{text}	0.40	Strongest discriminator. Semantic similarity captures service identity beyond lexical overlap. Calibration shows ST cosine $\geq$ 0.30 for 96% of verified same-contract pairs.
s_{cpv}	0.25	Structured and reliable when specific. Degrades gracefully through 4 levels. Not used as a hard gate because buyers sometimes reclassify.
s_{temp}	0.20	Rewards timing alignment without hard-excluding distant candidates (hard filters apply separately).
s_{buyer}	0.15	Constant within the loop; kept explicit so all weights sum to 1.0. Included for future extension to cross-buyer matching.

Table 11: Linking rate by method

Method	Linked	Denominator	Rate
Baseline (Jaccard + buyer × CPV groupby)	219	1,933	11.3%
TF-IDF cosine (intermediate)	279	1,100	25.4%
Sentence-Transformers (final)	**697**	**1,100**	**63.4%**

Table 12: Failure mode breakdown (403 unlinked eligible AO)

Failure reason	Count	Share	Description
`NO_TEMPORAL_PARTNER`	339	84.1%	No other AO from the same buyer fell in the ±12-month window. Likely structural (buyer issued only one contract in the period).
`TEXT_MISMATCH`	40	9.9%	Temporal candidates existed but all had $s_{\text{text}} < 0.20$. Text was too dissimilar even at the semantic level.
`CPV_MISMATCH`	24	6.0%	Temporal candidates existed, but the closest diagnostic candidate was in an incompatible CPV section. This label is a diagnostic warning, not a separate CPV hard gate in the final scoring rule.
Total unlinked	**403**	100%	

Table 13: Score calculation for pair (`15-31464` → `17-176560`)

Quantity	Description	Value
t_i	Source start date	2015-03-02
d_i	Declared duration	36 months
$\hat{e}_i$	Estimated end date ($t_i + d_i$)	2018-03-01
t_j	Candidate start date	2017-12-17
$\delta(i, j)$	Gap ($t_j - t_i$)	33.54 months
s_{text}	ST cosine of *objet* embeddings	0.6922
s_{cpv}	Source CPV missing ⇒ neutral	0.5000
s_{temp}	$1 - \lvert 33.54 - 36 \rvert/12 = 1 - 2.46/12$	0.7951
s_{buyer}	Same buyer group	1.0000
$S(i, j)$	$0.40(0.6922) + 0.25(0.5) + 0.20(0.7951) + 0.15(1.0)$	**0.7109**

Table 14: Source contract and candidate renewal

Element	Notation	Value
Source contract ID	i	`15-31464`
Source start date	t_i	2015-03-02
Declared duration	d_i	36 months
Estimated end date	$\hat{e}_i = t_i + d_i$	2018-03
Candidate contract ID	j	`17-176560`
Candidate start date	t_j	2017-12-17
Buyer match	$b_i = b_j$	Yes

10 Output Files Summary

The notebook produces four output files. The main modeling table is the renewal links file. The other files preserve candidate pairs, aggregate linking statistics, and diagnostic failure patterns for audit and sensitivity analysis.

Table 15: Output files produced by the notebook

File	Rows	Purpose
`boamp_renewal_links.csv`	1,100	**Modeling input.** One row per eligible AO; `event` column is the survival outcome.
`boamp_renewal_candidates.csv`	5,356	All filtered candidate pairs before best-match selection. Useful for threshold sensitivity analysis.
`boamp_linking_stats.csv`	1	Summary statistics: total AO, eligible, linked, linking rate.
`boamp_bias_report.csv`	71	Failure reason × CPV cross-tabulation. Diagnoses which categories are under-linked.

11 Next Steps (Phase 2)

Phase 1 delivers the event variable. The planned next phases are:

1. **Phase 2 — Survival model:** Fit a Cox proportional hazards or Kaplan–Meier model on `boamp_renewal_links.csv` to estimate time-to-renewal distributions and identify covariates (CPV, buyer type, contract size, procedure type) that predict early or late renewal.

2. **Phase 3 — Validation:** Manual annotation of a stratified sample of 50–100 linked pairs to estimate precision. Sensitivity analysis on the composite score threshold and temporal window.

3. **Phase 4 — DECP integration:** Merge with the DECP dataset (Déclaration des Données de la Commande Publique) to enrich with actual awarded amounts and contractor identities.

4. **Phase 5 — Production pipeline:** Operationalise as a monthly batch job on new BOAMP notices.